

Ragul MM

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SUMMARY

Software Engineer with 3 years of experience in Software testing, Python automation, and virtualization, specializing in Web applications. Skilled in creating and executing test plans based on customer requirements and automating test cases using Python, Selenium, MySQL, REST API and well proficiency in Postman Tool, AWS, GitHub, Mongo and worked with operating systems like Linux and Windows. And JavaScript as Intermediary. Collaborative team player committed to delivering high-quality software solutions.

SKILL SUMMARY:

Programming Languages:

- Python, Selenium, MySQL, REST API, Postman Tool, AWS, GitHub, Mongo
- JavaScript and HTML are Self-Taught

Automation Frameworks:

- Experience with Selenium Automation Framework and Pytest, Locust, K6, Jmeter

DevOps Tools:

- Basic of Kubernetes, and Intermediary in Jenkins

Version Control & SDLC Tools:

- Proficient in Git and Jira

Operating Systems:

- Experience with Linux and Windows

Kubernetes Skills

- Basic understanding of Kubernetes architecture and core concepts like Pods, Deployments, and Services.
 - Deployed containerized applications using Kubernetes in a development environment.
 - Experience with Kubernetes YAML files for defining resources like Deployments, Services, and ConfigMaps.
 - Familiar with kubectl for managing and interacting with Kubernetes clusters.
 - Used Minikube for local Kubernetes cluster setup and application testing.
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EDUCATION:

Madras Institute of Technology – Anna university
Bachelor of Engineering – Automobile | GPA – 80%

Chennai, India
2016 – 2020

SRV Hr Sec School
HSC | GPA – 87%

Rasipuram, India
Completed: 2016

SVN Hr.Sec School
SSLC | GPA – 96.4%

Erode, India
Completed: 2015

WORK EXPERIENCE:

Software Engineer At mieuPro Systems

Oct-2021- Nov-2024

Title : Ushur - Test Automation of AI-Based RPA Platform Software

Software : Python

Client : Ushur Inc., USA

Description :

- Ushur's RPA Platform is designed to automate monotonous workflows in Departments like Customer Support, HR, across various domains like Insurance, Banking
- Experienced with Manual Regression of Ushur specific modules like WebHook, OCR, Document Processor, Data Extractor, Credit Card, E-Mail, SMS, Phone Call, Incoming Email etc.,
- Experienced in REST API testing using Postman tool
- Extensively worked in testing the Platform's Language Processing algorithms such as,
 - Keyword, word and Doc vector combination
 - Keyword basis
 - Doc vector basis
 - Support vector classifier
 - Simple Neural Network
- Tested the above algorithms with different kind of datasets for various confidence levels
- Performed load testing using python by uploading multiple datasets to the system and evaluating the capacity of the system

Title : Elastic Kubernetes Service (EKS) Testing

Software : Cluster management, kubectl, Kubernetes API, JMeter, Locust, k6, Postman, REST Assured, Jenkins, Python, Git, GitHub, AWS

Client : Ushur

Description :

- **EKS Testing:** Developed and executed performance and functional test strategies for applications deployed on EKS. Utilized JMeter and k6 for load testing to identify and mitigate performance bottlenecks. Integrated EKS testing into CI/CD pipelines using Jenkins.

- **Performance Testing:** Designed and executed performance tests for Kubernetes clusters using JMeter and k6. Analyzed results to identify bottlenecks and optimize cluster configurations
- **Functional Testing:** Implemented functional testing strategies for containerized applications using REST APIs and performed functional testing of microservices deployed on EKS.
- **Monitoring & Optimization:** Set up monitoring and alerting using AWS CloudWatch. Analyzed performance metrics and logs to optimize EKS clusters and application performance.
- **Collaboration:** Coordinated with DevOps and development teams to define testing requirements and troubleshoot issues. Provided actionable insights and recommendations to improve application stability and performance.
- **Documentation:** Created detailed test plans, cases, and reports. Documented testing procedures and outcomes to ensure clear communication and effective tracking of issues and resolutions.

Title : Online Product Order Automation using Selenium & Python

Software : Python, Selenium WebDriver, PyTest, Allure Reports, POM, MySQL, Jenkins

Description :

- Developed an automation framework to test an e-commerce platform for ordering vegetables. Automated end-to-end scenarios, including login, searching for vegetables, adding items to the cart, applying discounts, and order confirmation.

Key Contributions:

- Designed a robust test automation framework using Selenium with Python following the Page Object Model (POM).
- Automated UI interactions for product search, cart management, and checkout flow.
- Implemented data-driven testing with MySQL to validate product details and pricing.
- Integrated PyTest and Allure Reports for structured reporting and test execution tracking.
- Developed test scripts to handle dynamic elements, pop-ups, and AJAX-based loading.
- Set up Jenkins for CI/CD automation, triggering test runs on every code commit.
- Used headless browser execution to optimize test runs for CI/CD pipelines.

Title : API Load Testing using Locust Framework

Software : Python, Locust, Requests Library, AWS EC2 (for distributed load testing)

Client : Ushur

Description : Designed and executed load tests for REST APIs using **Locust**, simulating real-world traffic to evaluate system performance under various loads.

Key Contributions:

- Developed **Locust test scripts** to simulate thousands of concurrent users performing API requests.
- Implemented **user behavior models** to mimic real-world API usage patterns.
- Configured **ramping-up scenarios** to analyze system stability under increasing loads.
- Integrated **distributed load testing** using multiple Locust workers on **AWS EC2**.
- Visualized key performance metrics (**response time, throughput, failure rate**) in **Zabbix** as the backend.
- Identified **API bottlenecks** and worked with the development team to optimize response times.

Title : Performance testing

Software : K6 (JS), Locust, RestApi, AWS, Redis

Client : Ushur

Description :

- **Script Developing:** Developed K6 script to test their Web application hosted in AWS focussing on the load,durability taken by the system
- **Performance Testing:** Performed performance testing on Web application using K6. Identified bottlenecks and optimized query performance to improve report generation times.

Collaboration: Worked closely with data engineers, Python developers, and stakeholders to understand requirements and testing criteria. Provided feedback on data discrepancies and suggested improvements.